

Advanced Data Communication.

Principles and Applications of Satellite.

STUDENT ID: PUIS/24210049. (ISE).

INTRODUCTION TO SATELLITE COMMUNICATIONS

On earth-space communications at these frequencies. ATS-5 was designed to operate as a gravity gradient stabilized satellite, unlike the earlier spin-stabilized ATS-1 and -3 satellites. It was successfully launched in August 1969 into synchronous orbit, but the gravity stabilization boom could not be deployed because of the satellite's spin condition. ATS-5 was placed into a spin-stabilized condition, resulting in the satellite antennas sweeping the earth once every 860 milliseconds. Most of the communications experiments performed with limited success in this unexpected 'pulsed' operation mode. The 15.3 GHz millimeter wave experiment downlink, however, was able to function well, after modifications to the ground terminal receivers, and extensive propagation data were accumulated at over a dozen locations in the United States and Canada.

ANIK A

ANIK A (initially called ANIK I), launched in November 1972 by NASA for Telsat Canada, was the first domestic commercial communications satellite. Two later ANIK As were launched in April 1973 and May 1975. The satellites, built by Hughes Aircraft Company, operated at C-band and had 12 transponders, each 36 MHz wide. The primary services provided were television distribution, SCPC (single channel per carrier) voice, and data services. The transmit power was 5 watts, with a single beam covering most of Canada and the northern United States. The antenna pattern for ANIK A was optimized for Canada, however sufficient coverage of the northern US was available to allow leased service by US communications operators for domestic operations prior to the availability of US satellites. The ANIK A series continued in service until 1985, when ANIK D satellites replaced them.

ATS-6

ATS-6, the second generation of NASA's Applications Technology Satellite program, provided major advancements in communications satellite technology and in new applications demonstrations. ATS-6 consisted of a 9 m diameter deployed parabolic antenna, earth viewing module, two sun-seeking

solar arrays, and the supporting structures. It was launched in May 1974 and positioned at 94° W longitude, where it remained for one year. In July 1975 it was moved to 35° E longitude for instructional television experiments to India. After one year it was again relocated to 140° W longitude and used for several experimental programs until it was moved out of synchronous orbit in 1979. ATS-6 had eight communications and propagation experiments that covered a frequency range from 860MHz to 30GHz. The communications subsystems on ATS-6 included four receivers: 1650 MHz (L-band), 2253 MHz (S-band), 5925–6425 (C-band), and 13/18GHz (K-band). Transmitter frequencies were: 860MHz (L-band), 2063MHz (S-band), 3953–4153MHz (C-band), and 20/30GHz (Ka-band). The ATS-6 provided cross-strapping at Intermediate Frequency (IF) between any receiver to any transmitter (except for the 13/18 GHz receiver, which operated with a 4150 MHz transmitter only), allowing a wide range of communications modes. Major experiments were:

- Position Location and Aircraft Communication Experiment (PLACE), which consisted of voice and digital data transmissions and four-tone ranging for aircraft position location. The system allowed multiple access voice from up to 100 aircraft, operating in 10 KHz channels.
- Satellite Instructional Television Experiment (SITE), a cooperative experiment between NASA and the government of India to demonstrate direct broadcast satellite television for

EARLY HISTORY OF SATELLITE COMMUNICATIONS

Instructional purposes. Satellite signals at 860 MHz were received at over 2000 villages, using simple 3 m parabolic antennas.

1. Television Relay Using Small Terminals (TRUST), which evaluated hardware and system performance for 860 MHz satellite broadcast television, using the same general configuration as the SITE.

2. Health Education Experiment (HEW), which provided satellite distribution of educational and medical programming, primarily to Alaska, the Rocky Mountain states, and Appalachia. Two independent steerable beams, operating through the 9 m reflector, were available, with the uplink at C-band (5950 MHz) and the downlink at S-band (2750 and 2760 MHz).

3. Radio Frequency Interference Experiment (RFI), which monitored the 5925 to 6425 MHz band with a sensitive on-board receiver to measure and map radio frequency interference sources in the continental United States (CONUS). The minimum detectable source EIRP was 10 dBW, with a frequency resolution of 10 kHz.

NASA Millimeter Wave Propagation Experiment, which was designed to provide information on the communications and propagation characteristics of the atmosphere at 20 and 30 GHz (Ka - band). Two modes of operation were available:

- i) downlink beacons at 20 and 30 GHz for measurement of rain attenuation, atmospheric absorption, and other effects; and
 - ii) a communications mode, with a C-band (6 GHz) uplink, and simultaneous downlinks at 20, 30, and 4 GHz, for the evaluation of millimeter wave communications in a 40 MHz bandwidth. Extensive measurements were obtained in the US and Europe, providing the first detailed information on Ka -band satellite communications performance.
4. COMSAT Millimeter Wave Experiment, which consisted of 39 uplinks, 15 at 13.19GHz, and 24 at 17.79 GHz, received by ATS-6 and retransmitted to the ground at C-band (4150 MHz). About one year of measurements were accumulated on rain attenuation statistics, joint probability distributions, and required rain margins, for links operating at K-band.

The accomplishments of ATS-6 have been extensively documented, and have provided a wide range of valuable design and performance information for virtually every application implemented in current satellite communications systems.

CTS

The Communications Technology Satellite (CTS) was a joint program between NASA and the Canadian Department of Communications, to evaluate high power satellite technology applicable to broadcast satellite service (BSS) applications at Ku -band. A 12 GHz, 200-watt output TWT on CTS, provided by NASA, allowed reception of television and two-way voice with small (120 cm diameter) ground terminal antennas. A continuously operating 11.7 GHz propagation beacon was also included, and long-term (36 month) propagation statistics were developed for several locations in the United States. CTS was launched in January 1976, and provided extensive experimental tests and demonstrations in the US and Canada until operations ended in November 1979.

Three important events helped to shape the direction and speed of satellite communications development in its early years:

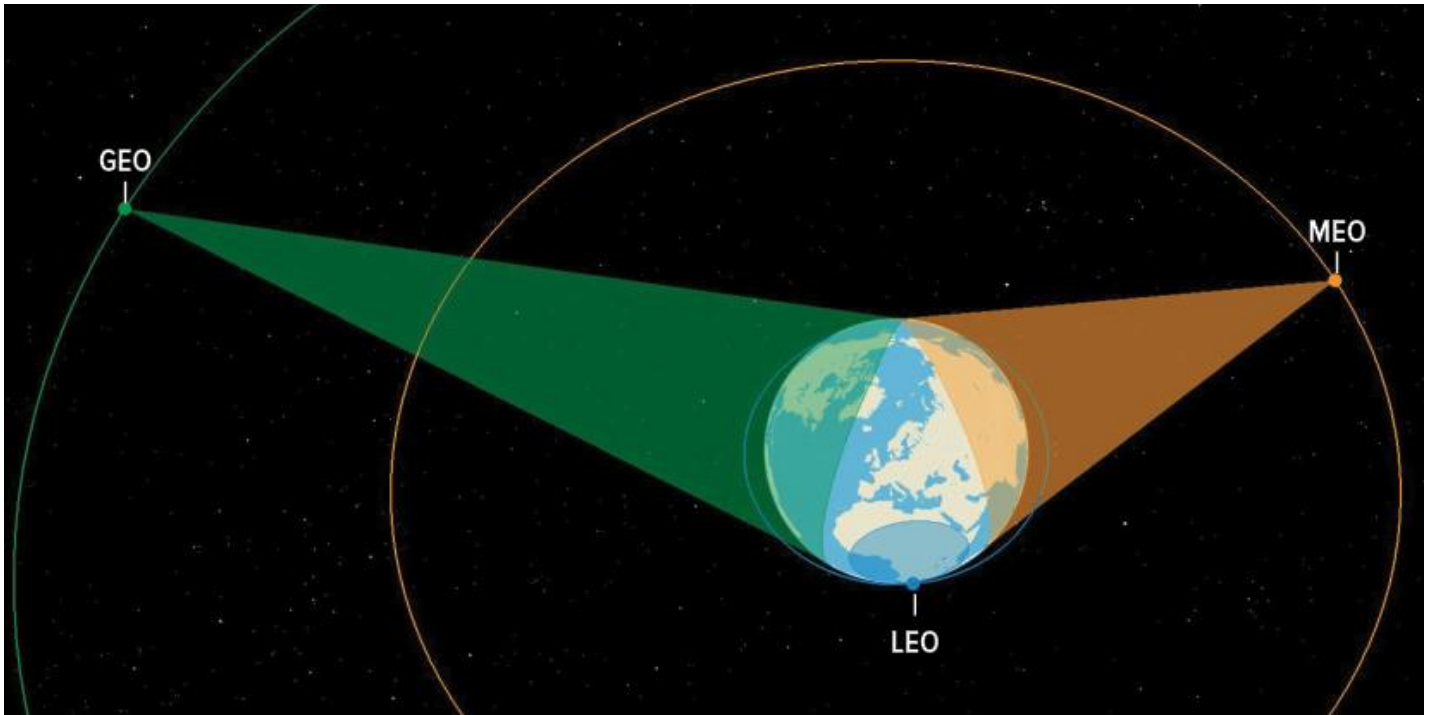
5. United Nations Initiative of 1961 – This initiative stated that ‘communications by means of satellite should be available to the nations of the world as soon as practicable.

A Brief Introduction to Low Earth Orbit (LEO) Satellites

What Are LEO Satellites?

Satellite technologies are divided into three categories based on the orbital range they occupy: Geostationary Earth Orbit (GEO), Medium Earth Orbit (MEO), and Low Earth Orbit (LEO).¹ LEO satellites operate in the closest range to Earth, orbiting between 300 and 2,000 kilometers above the planet’s surface. At this altitude, latency (or lag time) is reduced, thus enabling LEO satellites to provide near-real-time communications with less signal power for transmission.² This allows LEO satellites to be smaller in size than those in MEO or GEO, reducing the cost to develop and deploy them.

While LEOs can provide enhanced satellite-communication services due to their proximity to Earth’s surface, they do come with tradeoffs. LEOs have smaller coverage areas than MEOs and GEOs, and they must be constantly in motion to offset the pull of gravity and remain on their orbital path. Due to their proximity, movement, and small size, a “constellation” of hundreds or thousands of LEOs is typically needed to provide consistent service to an area. Additionally, LEO satellites, which complete an orbit approximately every 90 minutes, are fuel-intensive to operate and prone to atmospheric drag, which degrades satellites over time. As a result, the lifespan of a typical LEO ranges from seven to 10 years.



This figure shows the orbital ranges of three different kinds of satellites: Geostationary Earth Orbit (GEO), Medium Earth Orbit (MEO), and Low Earth Orbit (LEO). LEO satellites operate in the closest range to Earth, 300–2000km above the planet’s surface.

How Do LEO Satellites Work?

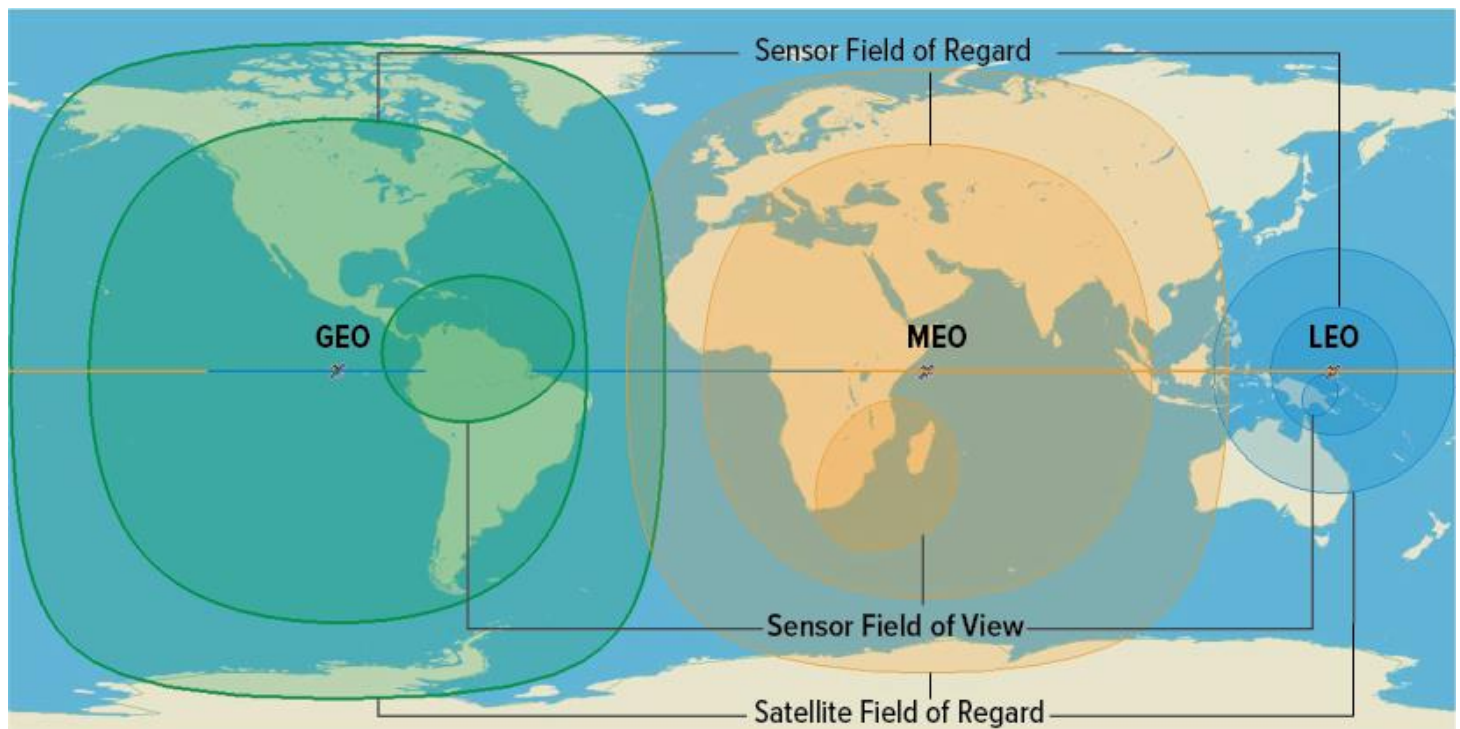
LEO satellite systems consist of three main components: the satellite constellation, user terminals, and ground stations. To establish a connection, terminals and ground stations must have clear lines of sight to satellites. Since LEO satellites are in constant movement, a constellation of hundreds or thousands of LEO satellites is typically needed to provide consistent service to an area, with user terminals and ground stations continually switching among different satellites within the constellation to maintain a user’s connection. While LEO satellites generally need to be within range of a ground station to send or receive data, some satellites also use inter-satellite links to share data among themselves until the data reaches a satellite within range of a ground station. Advancements in inter-satellite links, such as laser-based technologies, help LEO satellites provide reliable connectivity, even in areas far from a ground station.

Satellites use spectrum radio frequencies that transmit wireless signals to send (downlink) and receive (uplink) information from user terminals and ground stations. Since available spectrum is limited and

in most cases shared among operators, it is regulated nationally and internationally to promote global harmonization and to avoid interference or signal disruptions.

The International Telecommunication Union (ITU), an organization within the United Nations, allocates specific frequency bands for satellite communications with a variety of propagation characteristics and coordinates the registration of satellite-frequency assignments and (for the geosynchronous GEOs) their associated orbital positions. Currently, the most commonly used bands for connectivity-oriented LEO satellites are the Ku- (12–18 GHz) and Ka- (26.5–40 GHz) bands.

Satellite operators obtain licenses for spectrum use from their national regulators, who are responsible for ensuring registered satellite operators follow international guidelines, as well as any specific national regulations. Filing a registered satellite operator with the ITU is a multistep process designed to identify and mitigate any potential conflicts between a planned system's orbital characteristics, frequency bands, and intended service areas within an existing system. As such, once certain spectrum bands are in use, new entrants must design their systems around existing operators.

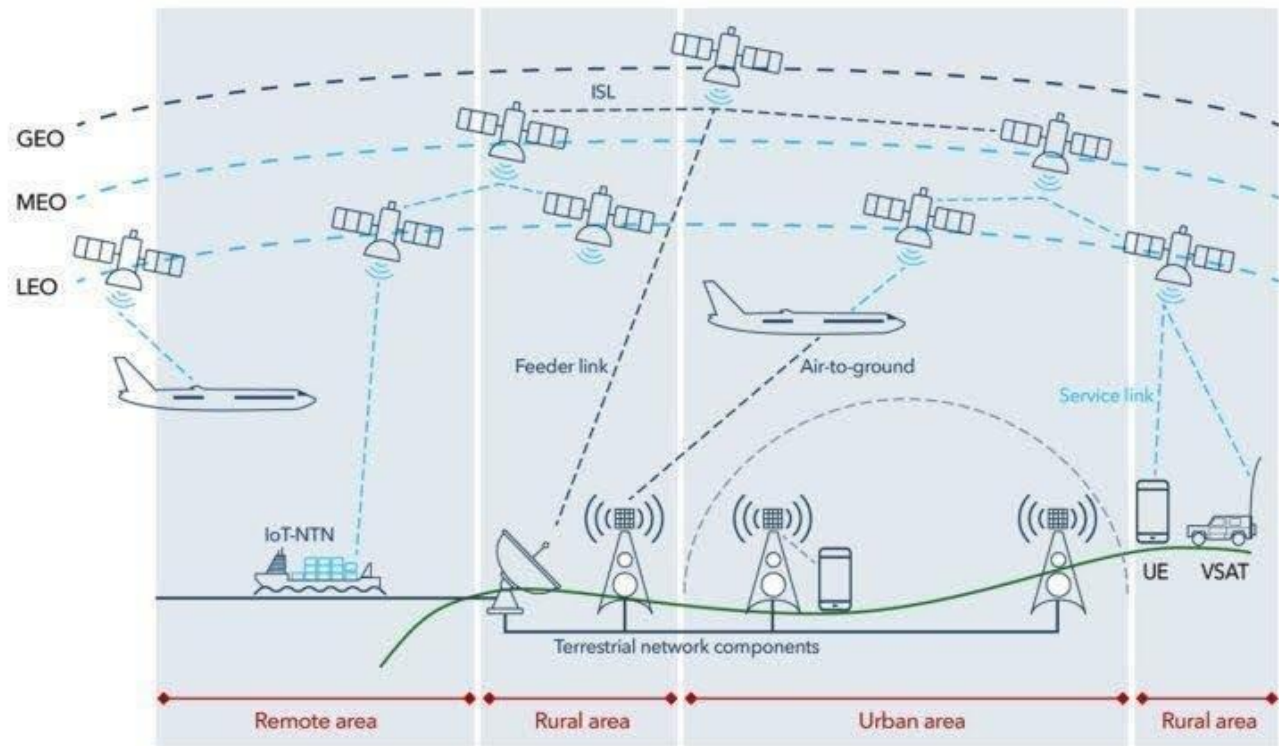


For communication satellites, coverage areas are dictated by proximity to Earth: A GEO can cover half the globe, but with slower speeds and longer lag times. LEOs offer high-speed, low-latency service, but require more satellites to cover large areas.

The Promise of LEO Satellites: Ubiquitous Connection and Versatility

Due to their global coverage and proximity to Earth, LEO satellites can offer a wide range of services as both an alternative and complement to existing connectivity services. LEO systems can provide broadband connectivity to residential, community, and enterprise customers, along with other services such as:

1. **Vehicles and platforms in motion:** In June 2022, the Federal Communications Commission (FCC) approved SpaceX's Starlink and Kepler Communications for earth stations in motion (moving or transportable ground terminals), allowing the use of LEO satellites to provide connectivity for vehicles in motion—including cars, trucks, ships, and planes.
2. **Areas experiencing natural disasters and conflict:** LEO satellite connections are being used in combination with GEO satellites to respond to natural disasters and connect areas experiencing conflict, such as Ukraine and Gaza, where existing infrastructure is damaged.
3. **Enterprise Internet of Things (IoT):** LEO satellites can offer enterprise connectivity services, such as providing backhaul to increase network coverage, enhance cloud storage, and provide support for edge computing. LEOs can also provide connectivity for widespread IoT devices and machine-to-machine communications, from asset tracking and remote monitoring to delivery drones and robotics.
4. **Direct-to-device mobile service:** The FCC's 2024 decision to authorize "supplemental coverage from space" allows LEO (and GEO) satellites to transmit directly to devices (D2D) on select mobile-carrier spectrum bands without the need for a ground station.¹¹ Starlink and T-Mobile have been authorized to provide this service, initially limited to texting. Similarly, LEO operators can use Mobile Satellite Service spectrum bands to transmit directly to mobile devices without an arrangement with a terrestrial mobile carrier, such as Apple's partnership with Globalstar, to transmit directly to iPhones and enable texting in even remote locations.



This figure shows the components of non-terrestrial networks: the satellite constellation, user terminals, and ground stations that backhaul data traffic to the internet. It also demonstrates use cases for satellites occupying different orbital ranges.

From GEO to LEO Satellites: A New Space Race

Global satellite communications first began in the 1960s with GEO and MEO satellites. In 1962, AT&T, with help from NASA, launched Telstar 1 a medium orbit satellite that enabled transatlantic television transmission from the United States to the United Kingdom and France. In 1964, Syncom 3, developed by Hughes Aircraft Co., became the first satellite in geostationary orbit, and transmitted live coverage of the 1964 Tokyo Olympics to one-third of the globe. From these initial launches until the 1990s, GEO and MEO satellites dominated the satellite-communications industry, delivering telephone and broadcast radio and television services across the globe.

In 1998, both Iridium and Globalstar launched LEO satellite mobile-communications services. High costs, limited market share, and long timelines to bring networks online, however, often meant the LEO networks of the 1990s failed to gain competitive viability relative to other terrestrial-service

providers. In 2003, Eutelsat would launch e-BIRD, the first geostationary satellite designed to deliver broadband.

Despite previous false starts, advances in satellite and launch technologies over the past decade have lowered the cost to bring LEO constellations online and to improve performance, thereby renewing the promise for future LEO-based communications and ushering in a new space race. By 2024 there were an estimated 10,000 active satellites in orbit, up tenfold from 1,000 satellites in 2014.

With operators around the world filing with the ITU to bring hundreds of thousands of new satellites online in the coming years, Goldman Sachs Research forecasts as many as 70,000 LEO satellites to be launched within the next five years. To deter early filers from “warehousing” spectrum or reserving it for future use, satellite operators must deploy systems within seven years of the ITU receiving its request or their claim expires. For LEO satellites, operators are required to deploy 10 percent of a planned constellation within two years, 50 percent within five years, and total deployment within seven years to maintain their spectrum authorization.

The LEO satellite field is occupied by only a small number of companies, with SpaceX’s Starlink dominating satellites in orbit. One estimate found that Starlink operated roughly 60 percent of 10,000 functioning satellites as of 2024. As of July 2025, Starlink reports having more than 7,800 satellites in orbit. Starlink originally planned to launch 12,000 satellites for its constellation, but later expanded the project to 42,000 satellites. Other companies—including Eutelsat’s OneWeb (operating 630 satellites), Telesat (operating 198 satellites), Amazon’s Project Kuiper (launched 129 of its planned 3,000 satellite constellation as of September 2025), and Iridium Communications (operating 80 satellites) are working to carve out market share for LEO-based connectivity for residential, enterprise, and government consumers.

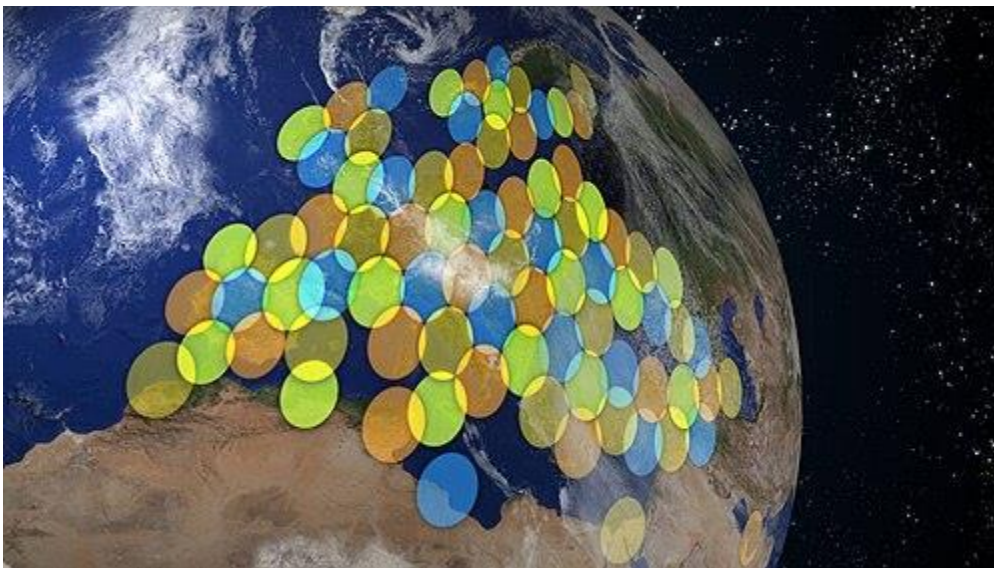
At the same time, many countries are exploring opportunities to advance and invest in the commercial LEO sector. Canada, China, and the EU are developing and heavily investing in national LEO satellite systems. The Canadian government provided Telesat Lightspeed a C\$2.14 billion loan to develop and operate a LEO network to advance national connectivity and defense. In December 2024, the European Commission announced a contract to build a multi-orbital constellation of 290 satellites for its Infrastructure for Resilience, Interconnectivity and Security by Satellite (IRIS²) project. In China, two megaconstellation projects Guowang and Qianfan have plans to launch 13,000 and 14,000 LEO satellites respectively. As the field of LEO satellite connectivity matures, the orbital range is expected

to become increasingly crowded, limiting orbital and spectrum availability while increasing space traffic and debris.

While LEO-based solutions are on the rise, some are already pushing for a new era of the final frontier: multi-orbit satellite connectivity. By integrating satellite service from multiple orbits, advocates hope to dynamically route service and capacity needs between GEOs, MEOs, and LEOs to provide reliable, flexible, and quality service for consumers. Developing multi-orbit service configurations depends, however, on first establishing a robust LEO satellite ecosystem.

High-throughput satellite.

A **high-throughput satellite (HTS)** is a communications satellite which provides more throughput than a classic fixed service satellite (FSS). An HTS provides at least twice, though usually 20 times or more, throughput for the same amount of allocated orbital spectrum, thus significantly reducing cost-per-bit. ViaSat-1 and EchoStar XVII (also known as Jupiter-1) provide more than 100 Gbit/s of capacity, which is more than 100 times the capacity offered by a conventional FSS satellite. When it was launched in October 2011, ViaSat-1 had more capacity (140 Gbit/s) than all other commercial communications satellites over North America combined.



KA-SAT coverage over Europe showing frequency reuse by different colors

List of high-throughput satellites

- Anik F2 (July 2004)
- Thaicom 4 (IPSTAR) (August 2005)
- Spaceway-3 (August 2007)
- WINDS (February 2008)
- KA-SAT (December 2010)
- Yahsat 1A (April 2011)
- ViaSat-1 (October 2011)
- Yahsat 1B (April 2012)
- EchoStar XVII (July 2012)
- HYLAS 2 (July 2012)
- Astra 2E (September 2013)
- O3b satellite constellation (2013-2014)
- Inmarsat Global Xpress constellation (2013-2015)
- Sky Muster 1 (NBN Co-1A) (30 September 2015)
- Badr-7 for TRIO Connect (November 2015)
- Intelsat 29e (2016)
- Intelsat 33e (2016)
- Sky Muster 2 (NBN Co-1B) (5 October 2016)
- SGDC (4 May 2017)
- SES-15 (May 2017)
- ViaSat-2 (June 2017)
- Intelsat 32e (2017)
- Intelsat 37e (2017)
- Intelsat 35e (2017)
- Eutelsat 172B (2017)
- GSAT-19 (2017)
- Shijian 13 (12 April 2017)
- SES-14 (January 2018)
- Yahsat-3 (25 January 2018)
- SES-12 (June 2018)
- GSAT-29 (14 November 2018)
- GSAT-11 (5 December 2018)
- Nusantara Satu (22 February 2019)
- Kacific-1 (17 December 2019)
- Eutelsat Konnect (16 January 2020)
- SES-17 (24 October 2021)
- Eutelsat Konnect VHTS (07 September 2022)
- O3b mPOWER satellite constellation (16 December 2022)
- ChinaSat 26 (23 February 2023)

Advantages and Disadvantages of Satellite Communication

Satellite communication is a form of wireless communication that transmits and receives information between points on Earth through artificial satellites orbiting the planet. This technology has become

essential for television broadcasting, weather forecasting, navigation, military communication, and global connectivity, especially in remote areas where terrestrial infrastructure is impractical.

What is Satellite Communication?

Satellite communication uses artificial satellites positioned in orbit around Earth to relay signals between ground stations. The satellite acts as a repeater, receiving signals from Earth, amplifying them, and retransmitting them back to different locations on the surface.

This system enables communication across vast distances and challenging terrains including oceans, mountains, and remote regions where traditional terrestrial networks are not feasible or cost-effective to deploy.

Satellite Communication System
Earth Satellite Station A Station B
Uplink Downlink

Advantages of Satellite Communication

Global Coverage

Satellites can provide communication services to remote and inaccessible areas including mountains, deserts, and ocean regions where terrestrial infrastructure deployment is impossible or economically unviable.

High-Speed Data Transmission

Modern satellites support high-bandwidth applications including video conferencing, internet connectivity, and real-time data transfer, making them suitable for demanding communication requirements.

Reliability and Availability

Satellite communication is less susceptible to natural disasters, earthquakes, or infrastructure damage that commonly affects terrestrial networks, ensuring consistent service availability.

Scalability and Flexibility

Additional satellites can be deployed to increase capacity and coverage area, allowing communication systems to adapt to growing demand and changing requirements.

Disadvantages of Satellite Communication

High Initial and Operational Costs

Satellite systems require substantial investment for manufacturing, launching, and maintaining satellites in orbit, making them expensive compared to terrestrial alternatives.

Signal Latency

The long distance signals must travel to and from satellites creates noticeable delays, particularly problematic for real-time applications like voice calls and interactive services.

Signal Interference and Attenuation

Weather conditions, atmospheric interference, and solar activity can degrade signal quality or cause complete service interruption, affecting communication reliability.

Environmental Impact

Satellite launches consume significant resources and contribute to space debris, while end-of-life satellites pose orbital pollution challenges.

Key Applications

Application	Description
Television Broadcasting	Direct-to-home satellite TV services reaching global audiences
Navigation Systems	GPS and other positioning systems for transportation and mapping
Weather Monitoring	Meteorological data collection and climate observation
Emergency	Disaster response and rescue coordination

Application	Description
Communication	in affected areas
Military Operations	Secure communication channels for defense and intelligence

Conclusion.

Satellite communication offers unique advantages including global coverage and reliability, making it essential for applications where terrestrial networks are inadequate. However, considerations of cost, latency, and environmental impact must be weighed when selecting communication solutions for specific requirements.

Satellite vs. Terrestrial Communications: Pros and Cons

With the rapid development of technology, 5G has become the new buzzword in the telecommunications industry. It promises to bring a revolutionary change in the way we communicate and access information. However, the debate between satellite and terrestrial communication systems has been ongoing for decades.

In this blog post, we will compare the pros and cons of satellite and terrestrial communications.

Satellite Communications

Satellite communication is a form of wireless communication that uses artificial satellites to relay signals between different points on Earth. **Satellite communication solutions** have been used for decades to provide communication services in remote and hard-to-reach areas. Here are some of the pros and cons of using satellite communication for 5G and beyond.

Pros

Global Coverage: One of the major advantages of satellite communication is its global coverage. Satellites can cover the entire globe, including remote and rural areas, making it an ideal solution for providing communication services in hard-to-reach areas. It has even got **communication services optimised for the maritime sector**.

Low Infrastructure Cost: Another advantage of satellite communication is that it requires low infrastructure cost. Once the satellite is in orbit, it can provide communication services to a large area without requiring any additional infrastructure.

Resilience: Satellite communication is also more resilient than terrestrial communication systems. It is not affected by natural disasters such as hurricanes, earthquakes, and floods, making it a reliable option for emergency communication.

Cons

Latency: One of the major drawbacks of satellite communication is the latency issue. Due to the long distance between the satellite and the receiver, there is a delay in the signal transmission, which can affect the quality of communication.

Signal Interference: Another disadvantage of satellite communication is signal interference. Weather conditions, solar flares, and other environmental factors can affect the signal quality and cause interference. It is especially problematic for the **maritime satellite communication** sector.

Cost: Satellite communication is more expensive than terrestrial communication systems. The cost of launching and maintaining satellites is high, and this cost is passed on to the end-users.

Terrestrial Communications

Terrestrial communication systems use ground-based infrastructure to provide communication services. These systems include cellular networks, Wi-Fi, and fiber-optic networks. Here are some of the pros and cons of using terrestrial communication systems for 5G and beyond.

Pros

Low Latency: One of the major advantages of terrestrial communication systems is low latency. The distance between the transmitter and the receiver is short, which results in faster signal transmission and low latency.

High Speed: Terrestrial communication systems can provide high-speed communication services, making them ideal for data-intensive applications such as video streaming, gaming, and cloud computing.

Cost-effective: Terrestrial communication systems are cost-effective compared to satellite communication solutions. The cost of setting up and maintaining terrestrial infrastructure is low, and this cost is passed on to the end-users.

Cons

Limited Coverage: Terrestrial communication systems have limited coverage compared to satellite communication systems. They are not able to provide communication services in remote and rural areas where **satellite communication solutions** are the only option.

Vulnerable to Natural Disasters: Terrestrial communication systems are vulnerable to natural disasters such as hurricanes, earthquakes, and floods, which can damage the infrastructure and disrupt communication services.

Network Congestion: Terrestrial communication systems can experience network congestion during peak hours or in high-traffic areas. This can result in slow data speeds and poor communication quality.

In conclusion

The comparison between **satellite** and terrestrial communication systems for maritime communication services shows that both have their unique advantages and disadvantages. However, when it comes to reliable and efficient communication services in the middle of the ocean, satellite communication systems remain the primary choice for the maritime industry. Satellite communication systems provide global coverage and resilience, enabling communication services to be available wherever ships are located, regardless of distance.

With years of experience in the industry, IEC Telecom offers a range of reliable and efficient maritime satellite communication solutions tailored to meet the specific needs and requirements of each client. From voice communication to data communication and positioning and navigation services, IEC Telecom has the expertise to provide the best communication solutions for maritime operations.

In conclusion, for the most efficient and reliable maritime communication services, satellite communication systems remain the primary choice, and IEC Telecom is the leading provider in the UAE. Contact IEC Telecom today to learn more about their innovative and customized solutions for the maritime industry.

Some urgent Incident Response Challenges In Space And Aerospace.



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From communication latency and radiation-induced glitches to orbital debris and cyber vulnerabilities, incident response in space and aerospace environments brings unique and often high-stakes challenges. When things go wrong in orbit, no one can intervene in real time—distance, delay and the

sheer cost of assets turn every second into a critical window to prevent mission failure or the loss of vital infrastructure.

As satellites, spacecraft and supporting ground systems become foundational to global navigation, defense and commerce, the urgency of solving these problems grows. Below, members of Forbes Technology Council share some of the most challenging—and pressing—incident response hurdles in space and aerospace and explain why addressing them is vital to protecting innovation and infrastructure both on and beyond Earth.

1. Overcoming Distance And Latency

In space, you can't "roll a truck" to fix a compromised system, and latency and distance make real-time response nearly impossible. This is an urgent challenge to solve because satellites now underpin banking, logistics and defense. Without autonomous, resilient incident response, one breach in orbit cascades into global disruption. - Dan Haiem, AppMakers USA

2. Protecting GPS From Vulnerabilities And Jamming

U.S. military operations depend critically on GPS for positioning, navigation and timing—creating vulnerabilities adversaries can exploit. GPS satellites face direct threats, endangering conventional operations and nuclear deterrence. Quantum sensing now delivers jamming-resistant solutions: inertial sensors with optical or chip-scale atomic clocks and magnetic and gravity-based navigation systems. - Prineha (Pri) Narang, DCVC

3. Tracking And Predicting Orbital Debris

One unique incident response challenge in space is the inability to monitor and predict orbital debris accurately. As satellites operate, collisions can cause catastrophic failures. Developing real-time tracking systems with AI for predictive analytics is urgently needed to protect assets and ensure continuous operation of critical services like communications and navigation on Earth. - Roman Vinogradov, Improvado

4. Reducing Communication Latency

Aerospace faces a unique incident response challenge: communication latency and limited bandwidth with satellites and spacecraft. When anomalies strike, operators may have to wait for telemetry, risking

billion-dollar assets and critical services like GPS. The solution to this urgent problem is onboard AI autonomy, which enables real-time anomaly detection, triage and self-healing to safeguard infrastructure on Earth. - Raghu Para, Ford Motor Company

5. Managing Firmware Updates

Based on my experience evaluating and designing these systems, a unique challenge is managing firmware upgrades across a two-front war. Pushing a patch to a satellite is a gamble—a failure can brick the asset. Likewise, we must update remote devices that accelerate satellite communications on Earth. A failed push in either domain can sever vital infrastructure links. - Anil Pantangi, Capgemini America Inc.

6. Securing Hackable Infrastructure

Satellites power global telecom, internet, defense, disaster response and navigation systems. Yet ground stations and spacecraft links remain hackable, vulnerable to jamming, spoofing and hijacking. Incident response is uniquely difficult since there's no quick physical access to space assets. Securing these systems is urgently needed to safeguard defense systems, economies, global navigation, communication and crisis response. - Harikrishnan Muthukrishnan, Florida Blue

7. Preventing Collisions With Space Debris

Space debris can smash satellites, but agentic AI sentinels could help satellites dodge it autonomously by sensing threats and maneuvering in a blink. It sounds wild, but it's near—and it's urgent, as collisions risk GPS and communications blackouts on Earth. These AI guardians would keep orbits clear, saving our technology and future missions. - Durga Krishnamoorthy, Cognizant Technology Solutions

8. Responding Quickly To Cyber Or Space Weather Events

In space, a unique incident response challenge is latency and distance—you can't instantly "patch" a satellite like a server. When spacecraft face space weather or cyberattacks, delayed communication and limited intervention make response harder. Solving this is urgent: Satellites power GPS, communications and security. Protecting them safeguards critical infrastructure in orbit and on Earth. - Shelli Brunswick, SB Global LLC

9. Responding To Failures In Mega-Constellations

Mega-constellations (like Starlink, with its thousands of satellites) amplify incident response challenges. A fault in one can ripple through interlinked satellites, multiplying the damage. Traditional siloed response fails here. Coordinated, constellation-level containment is urgent, because a single mishap could disrupt global internet, navigation and emergency communication systems on Earth. - Jagadish Gokavarapu, Wissen Infotech

10. Preventing Orbital Debris Escalation

One overlooked challenge in space incident response is orbital debris escalation. A single mishandled anomaly can fragment assets into thousands of high-velocity debris pieces, threatening satellites for generations. Building “debris-aware” incident response is urgent to safeguard navigation, climate monitoring and national security in space and on Earth. - Anusha Nerella

Cyberspace Security – Low Earth Orbit



Introduction

As satellites become increasingly integral to global communication, navigation, and surveillance, targeting their vulnerabilities by malicious actors to disrupt services or gain sensitive information will be imminent to maintain cyberspace security.

Cybersecurity in cyberspace is a versatile field dedicated to safeguarding digital information and infrastructure from a spectrum of cyberthreats and attacks. Cyberspace focuses on protecting the integrity, confidentiality, and availability of satellite systems and their associated data from cyberthreats and attacks.

The importance of cybersecurity controls regarding cyberspace includes robust encryption protocols, secure communication channels, regular system updates, and comprehensive threat monitoring to mitigate risks.

This article will focus on low Earth orbit (LEO), acknowledge medium Earth orbit (MEO) and geostationary Earth orbit (GEO), and discuss the main differences between them (Figure 1).

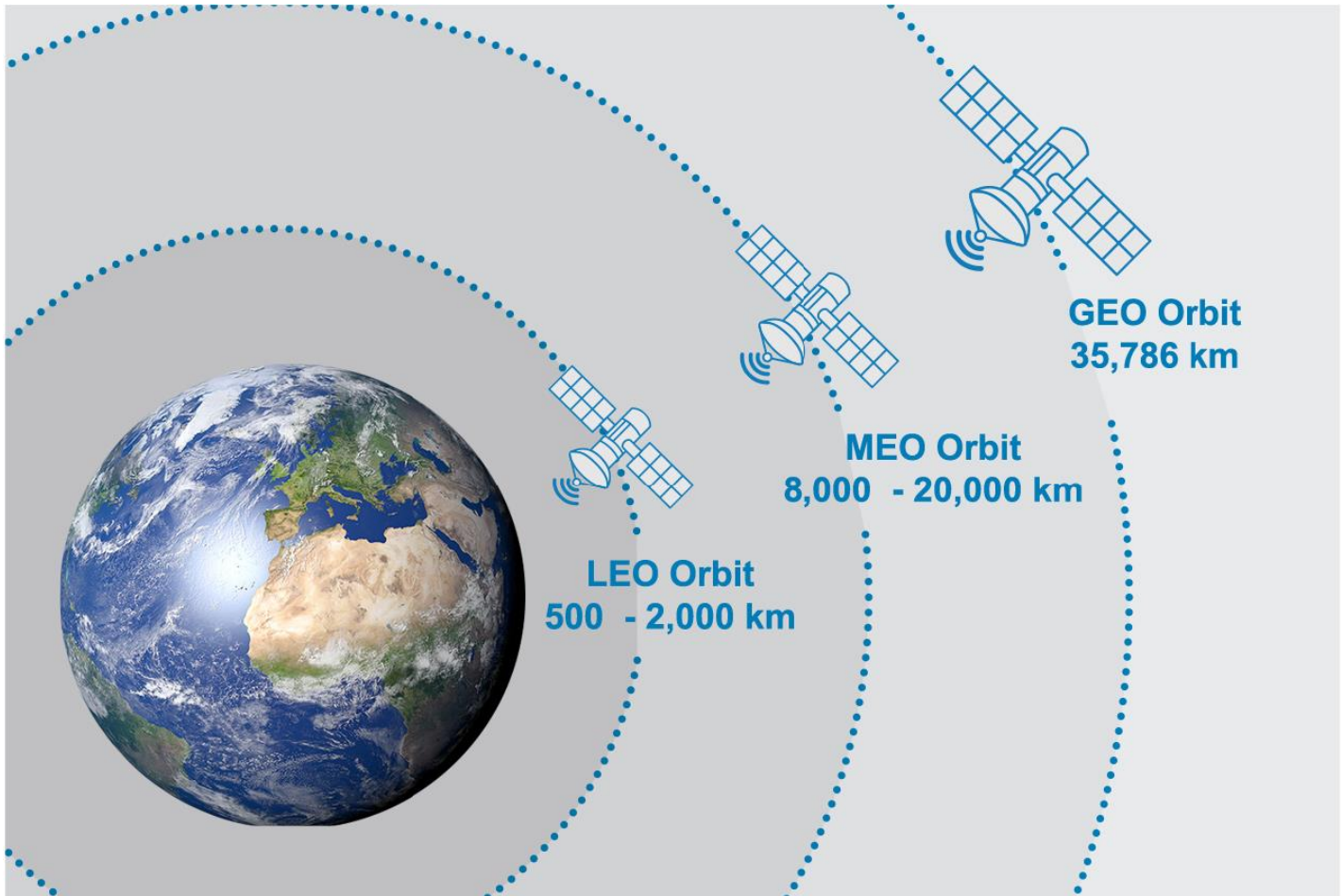


Figure 1. Classification of Satellite Orbits.

Exploring Various Types of Orbits

An orbit is the curved path that an object like a satellite in space takes around another object due to gravity. In space, there is no air and, therefore, no air friction. Because of this, gravity lets the satellite orbit around Earth with almost no further assistance [2].

For cybersecurity specialists, there will be some requirements to understand the basic terminologies regarding satellites' cyber safety and exploring their orbits, as the future of this country and its security depend on it.

As of July 2024, there were 6,281 Starlink satellites in orbit, with a potential to reach 42,000 satellites forming a mega constellation [3] (Figure 2).



Figure 2. Starlink V2 Satellites Launched From Space Launch Complex 40 (SLC-40) at Cape Canaveral Space Force Station in Florida.

LEO

LEOs will be the most accessible part of space for a broad range of operators. Even Amazon Web Services and Microsoft's Azure now provide Ground Station as a Service to enable communication with LEO satellites. LEO will initially host most space-based computing devices, which will require robust cybersecurity measures and controls, making them a primary target for adversaries.

Any space object that exists in an orbit below an altitude of 2,000 km above Earth is considered a LEO (Figure 1). A popular type of LEO satellites is known as "SmallSats," meaning small satellites [5]. Due to their proximity to Earth, these satellites are extensively utilized for a variety of applications, including telecommunications, Earth observation, and scientific research, leveraging their advantageous low-latency characteristics and enhanced resolution capabilities.

LEOs come in the following wide range of sizes and masses:

- Minisatellite: 100–180 kg

- Microsatellite: 10–100 kg
- Nanosatellite: 1–10 kg
- Picosatellite: 0.01–1 kg
- Femtosatellite: 0.001–0.01 kg

LEO SmallSats' cybersecurity controls are characterized as CubeSats. They are measured by units of U's, where 1U is the equivalent of $10 \times 10 \times 10$ cm [6].

MEO

MEO satellites are higher than the 2,000-km altitude and lower than the GEO satellites. They are primarily recognized for supporting regional coverage in global navigation satellite systems. A historical analysis of the progression of these satellites will provide a more nuanced and thorough understanding of their developing and current capabilities.

The United States initiated the deployment of its Global Positioning System (GPS) in early 1978, with the individual satellites designated as Navstar. It was not until 1982 that the Soviet Union launched its own counterpart, the Global Navigation Satellite System (GLONASS). The Chinese system, known as Beidou (meaning “compass”) was introduced later, with its first satellite launched in 2007 [7].

While MEO satellites compare in size to a standard refrigerator or a small car, the size of GEO satellites has significantly evolved since their inception, expanding from a mere few hundred kilograms to over six metric tons today.

GEO

GEO satellites typically range in size from about 1,000 kg (2,200 lb) to over 6,000 kg (13,200 lb) [8]. Their dimensions can vary widely, with some being as large as a car and others as large as a bus. Generally, they have large solar arrays and antennas, which can extend their overall size drastically.

Why Altitude Matters

Satellites traverse earth's orbit due to the intricate equilibrium between gravitational forces and their orbital velocity. Upon launch, a satellite reaches a precise velocity that permits it to perpetually descend toward Earth while advancing forward, thereby achieving a stable orbit.

Since SmallSats are within a certain altitude, they orbit Earth every 60 to 90 min. While MEOs orbit Earth between 8 to 15 hr, GEOs take 24 hr to orbit.

The higher a satellite orbit, the slower it moves. Since GEOs are equatorial orbits, they will be above the same spot over Earth at all times. This is extremely important, as it impacts the windows of communication with ground stations.

SmallSats orbit earth every 90 min, and the window in which they will be visible to their ground stations is only a few minutes. Because they are so close to Earth, they do not consume much energy to get their communication signals to the ground stations or receive instruction commands or a payload patching a vulnerability in their software [9].

The Cyberspace Challenge

When it comes to cyberattacks within cyberspace, there is always a motive and an intention.

Intention denotes the specific plan or goal an individual or a group aims to achieve with their actions, whereas motive encompasses the underlying reasons or drives that inspire those actions. Intentions are typically conscious, while motives can operate at both conscious and subconscious levels.

The cyber risk analysis regarding satellite systems is based on intention and motivation of threat actors and determination of the impact and probability of success undermining the satellite integrity and/or availability. The type of missions conducted by the targeted satellite will dictate the most effective vectors for cyber activity and the subsystems most susceptible to exploitation. These incorporated satellite mission types could be remote sensing or emitting types [10].

Regularly referred to as Earth observation satellites, sensing satellites are specialized satellites designed to collect data regarding the Earth's surface and atmosphere. Sensing is through electromagnetic spectrum sensors, radio waves, infrared radiation, and visible light, which are the same types of data commonly used in internet-based mapping applications or for weather forecasting and meteorological analysis and oceanography.

Emitting satellites are crucial for global communication systems and designed to transmit signals, data, or broadcasts back to Earth or to other satellites. These satellites facilitate the transmission of signals for satellite television services and help provide broadband connectivity to remote areas.

Emitting could also be used in overt or covert operations. Jamming or blocking signals is a classic example of an overt operation targeting other satellites’ sensors and performing a communication takeover. Covert operations like spoofing signals are similar to what is seen in electronic warfare.

Defending the Kármán Line

In cybersecurity, one of the most popular and effective cyber defense architectures is defence-in-depth (Figure 3), where there is perimeter security. Imagine this perimeter security is the Kármán line, which is the line used for regulatory purposes to differentiate between aircraft and spacecraft spreading a boundary 62 miles or 100 km above mean sea level that borders Earth’s atmosphere and the beginning of space [11].

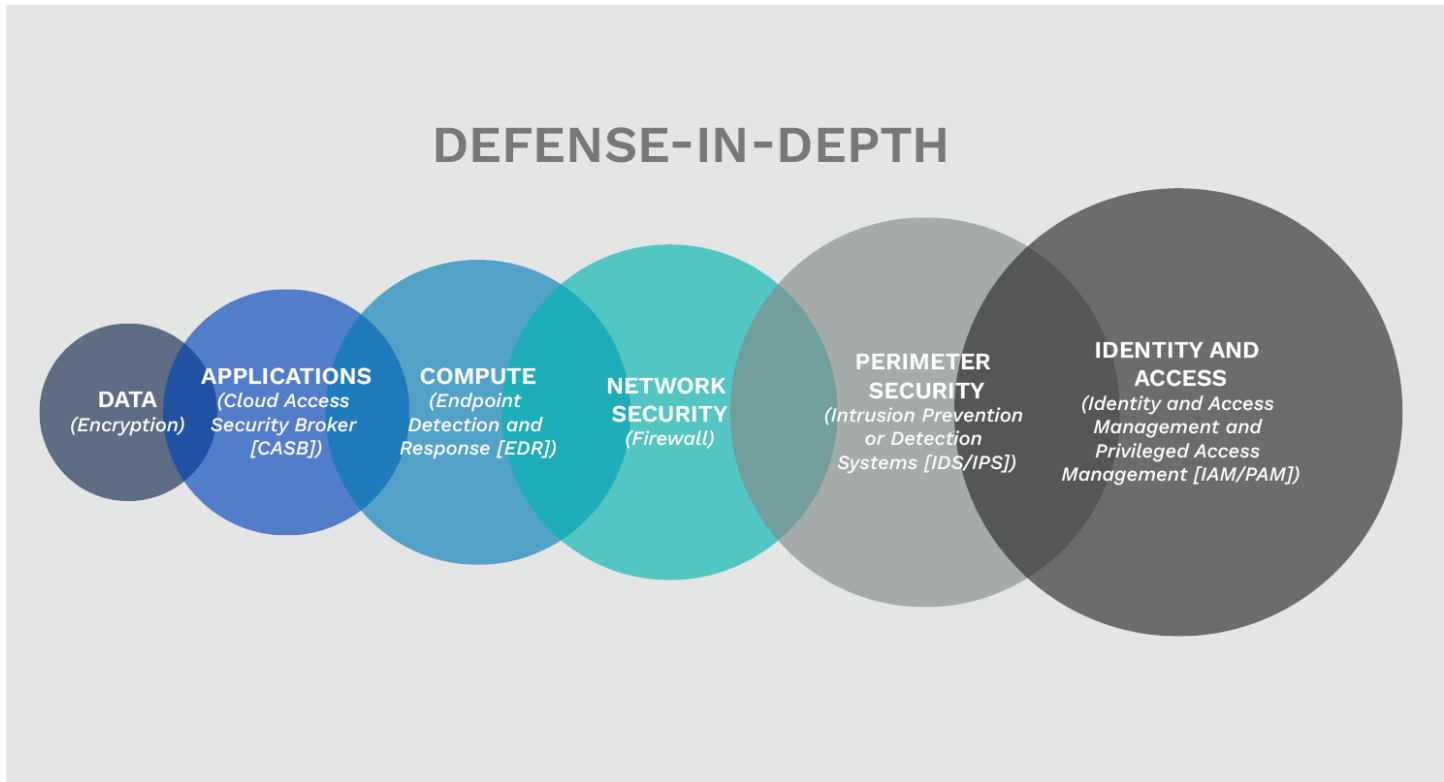


Figure 3. Defence-in-Depth Cybersecurity.

The simplest example of a space system involves a ground-based station device communicating with a space-based device “satellite” where both devices transmit and receive signals (Figure 4).



Figure 4. Satellite Communications Relay.

On the ground-base station, a software-defined radio (SDR) is responsible for receiving various signals like modulation, demodulation, filtering, and encoding and turning them into communications. Providing decryption of the communications stream passes it to a flight control computer running the software that communicates and controls the satellite used for keeping track of its flight operation and trajectory. This provides greater versatility and efficiency in modern communication systems and electronic warfare.

An attacker with access to one or more satellites could potentially redirect these satellites to receive commands not only from the legitimate ground station but also from attacker-controlled ones. By configuring a compromised satellite to listen for and accept instructions from a rogue ground station, the threat actor would undermine the integrity of the communication system.

SDRs enable cyberattacks to compromise communications either from the ground or the satellite. Both ground stations and space vehicles commonly use SDRs to configure, transmit, and receive signals through their antennas. An attacker could exploit vulnerabilities in these SDRs to disrupt

communication streams, either by altering configurations or introducing gradual subtle degradation. This kind of attack might not result in an immediate communication shutdown that triggers a rapid response from operators. Instead, it could cause intermittent and unreliable communication between ground and space systems. Consequently, operators might redirect communications to alternative ground stations, affecting the coverage and effectiveness of the space vehicle or its network.

Inadequately implemented encryption jeopardizes confidentiality and creates a misleading sense of privacy and security for the communicating parties. This illusion of protection persists until the parties discover that the secure encryption has been compromised.

Ground-to-space communications encounter augmented risks pertaining to the resilience of encryption. Unlike wired or other mediums, these communications are continuously transmitted through air, making them susceptible to interception. Even though the data is encrypted, the constant and extensive transmission exposes the encryption to potential analysis. An attacker might exploit these frequent and large communication sessions to detect patterns and potentially break the encryption.

Cyberspace Mitigations and Controls

There are many safeguards and controls already incorporated in most satellites. These controls focus on redundancy, with some resilience embedded into their architecture.

A recent analysis conducted by a team of German researchers offers an insightful examination of the security vulnerabilities present in satellites currently orbiting Earth. The researchers, affiliated with Ruhr University Bochum and the Cisca Helmholtz Center for Information Security, scrutinized the software utilized by three small satellites and discovered significant deficiencies in basic protective measures. According to their findings, the satellites assessed exhibited vulnerabilities within their firmware, revealing that minimal security advancements from the past decade have permeated the space domain.

Notably, these satellites lack adequate safeguards concerning who can communicate with their systems and do not incorporate encryption protocols. The researchers suggested that such shortcomings could allow an adversary to seize control of a satellite, posing risks of collision with other objects.

The analysis identified six distinct types of security vulnerabilities across the three satellites, totaling 13 vulnerabilities. Among these are “unprotected telecommand interfaces,” which are critical for satellite operators on the ground to communicate with the spacecraft in orbit.

SmallSats and CubeSats

These nanosatellites exhibit heightened vulnerabilities to cyberattacks due to their low construction costs for commercial entities. The proliferation of thousands of satellites in constellations congests LEO, creating fertile ground for malicious actors, especially in conjunction with military satellite deployments. Compromising these satellites could lead to significant economic ramifications and even potential loss of lives.

For instance, by compromising the satellite’s navigation system, an attacker could cause failure in its docking maneuvers or alter its orbital trajectory, potentially redirecting it to face Earth rather than the Sun. This will be explored in the “RoSat Attack” section of this article.

Private Sector and Field-Programmable Gate Arrays (FPGAs)

The private sector has largely neglected cybersecurity, possibly due to a lack of awareness and compounded by the financial burden of adequately securing satellites against cyberthreats and the absence of regulatory frameworks.

Implementing advanced encryption methods like quantum encryption could substantially fortify the cybersecurity of satellites against these threats, coupled with deploying secure gateways reinforced by intrusion prevention systems.

While FPGAs are useful for monitoring systems and overseeing the logic necessary to keep them running with minimal interruption when individual components fail, a complete microcontroller watchdog with logic for failure handling can be implemented in a hardware description language [13]. Invoking or triggering the watchdog scripts or watchdog timer (WDT) by various situations will correct an error in navigation after a certain threshold.

WDT

A WDT is a system monitoring mechanism designed to detect and respond to failures or malfunctions in satellite software or hardware; it could be a lifesaver for SmallSats. The WDT operates by resetting

at regular intervals, requiring the system to reset within a specified timeframe to prevent errors or crashes. If threat actors were to disrupt the GPS system from a ground station, a WDT might eventually take over the navigation system, allowing the space system operators to regain control of the space vehicle.

Space Attack Research and Tactic Analysis (SPARTA)

SPARTA is one threat mitigation framework for space attacks. While the SPARTA matrix framework is used to illustrate some examples, there are also several other key initiatives and agencies focused on cyberspace security, including the Space Information Sharing and Analysis Center, the Defense Advanced Research Projects Agency, and the European Space Agency (ESA).

Addressing the information and communication barriers that hinder the identification and sharing of space-system tactic, techniques, and procedures, SPARTA provides unclassified information to space professionals about how spacecraft may be compromised via cyber and traditional counterspace means (Figure 5). The matrix defines and categorizes commonly identified activities that contribute to spacecraft compromises [16].

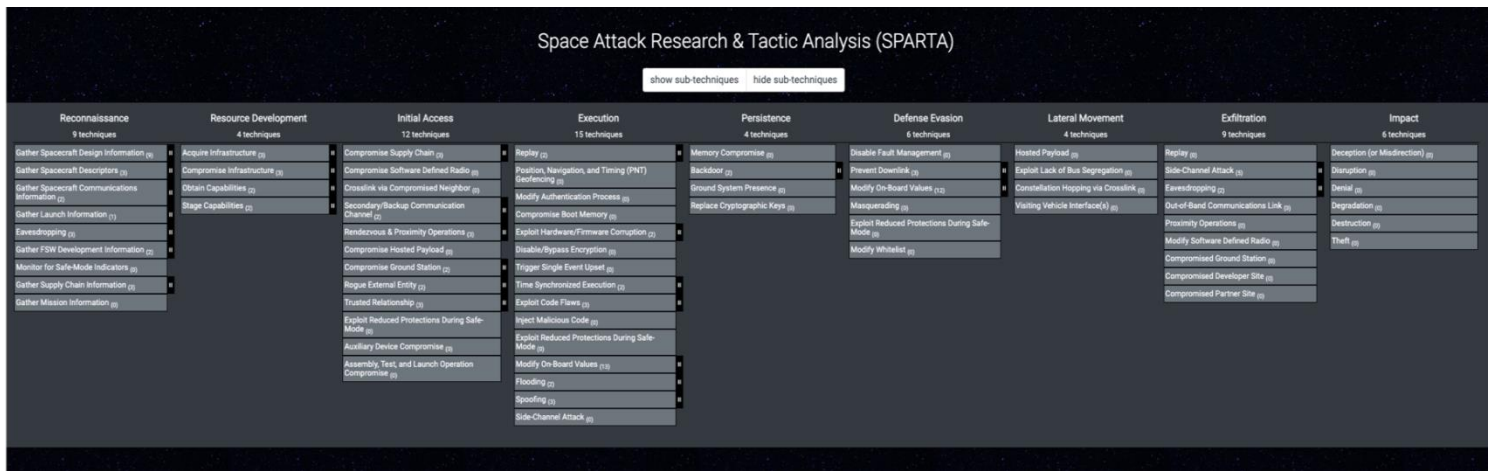


Figure 5. SPARTA Tactics and Techniques Matrix.

For example, the “Initial Access” for the “Compromise Ground System” technique is comprised of two subtechniques: “Compromising On-Orbit Update” and “Malicious Commanding via Valid GS” (Figure 6).

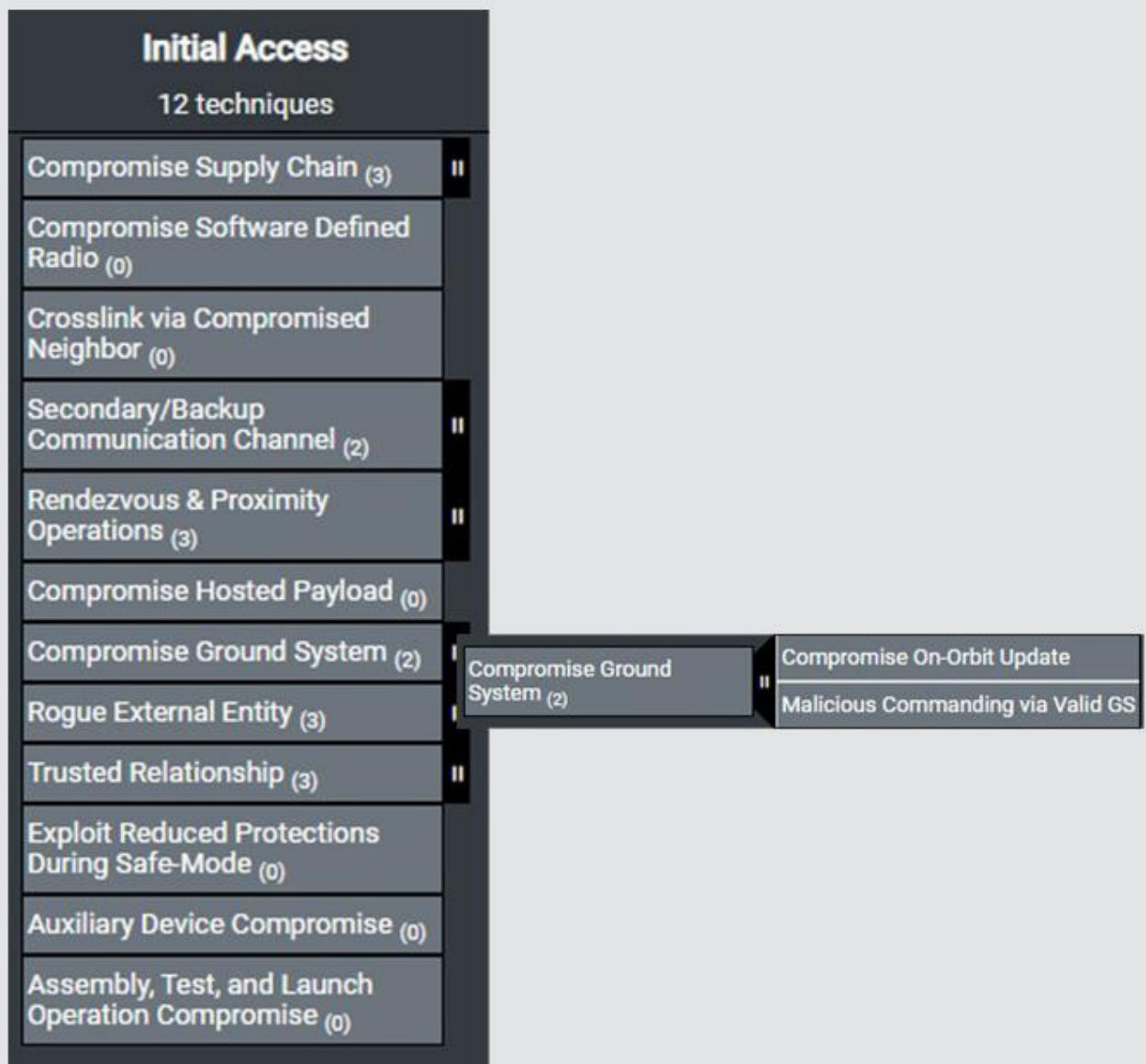


Figure 6. Initial Access Tactic – Compromised Ground System Technique.

Selecting Compromising On-Orbit Update

The mapping to NIST SP 800-53 Rev5, D3FEND, and ISO27001 makes it very accessible for cybersecurity teams to embed into their cyber programs and understand exactly which countermeasures they need to apply.

If these techniques (Figure 7) were applied properly, RoSat, the first cyberspace incident mentioned next, would have been easily avoided.

Home > Techniques > Compromise Ground System > Compromise On-Orbit Update

Compromise Ground System: Compromise On-Orbit Update

Threat actors may manipulate and modify on-orbit updates before they are sent to the target spacecraft. This attack can be done in a number of ways, including manipulation of source code, manipulating environment variables, on-board table/memory values, or replacing compiled versions with a malicious one.

Other Subtechniques of Compromise Ground System (2)

ID	Name
IA-0007.01	Compromise On-Orbit Update
IA-0007.02	Malicious Commanding via Valid GS

ID: IA-0007.01

Sub-technique of: IA-0007

Notional Risk (H | M | L): 24 | 21 | 17

Related Aerospace Threat IDs: SV-AC-1 | SV-IT-5 | SV-MA-7 | SV-SP-9

Related MITRE ATT&CK TTPs: T1195 | T1195.002

Related ESA SPACE-SHIELD TTPs:
No related Spaceshield TTPs

①Tactic: Initial Access

Created: 2022/10/19

Last Modified: 2024/02/29

Figure 7. Compromised Ground System Technique and Subtechniques.

Cyberspace Incidents

The ability to disable or destroy satellites through cyber exploitation is no longer theoretical—it is a present and growing threat. As space-based infrastructure becomes increasingly vital for communication, navigation, surveillance, and national security, cyberattacks targeting these assets represent one of the most disruptive and potentially devastating threats of the 21st century. Such attacks can compromise sensitive data, disable critical systems, or even render entire satellite constellations inoperable, with far-reaching consequences for both civilian and military operations.

RoSat Attack

In late 1998, a joint German and U.S. X-ray sensor satellite known as Röntgensatellit (RoSat) was compromised. The compromise involved a foreign threat actor gaining access to Goddard Space Flight Center using social engineering techniques, combined with an inadequately configured file transfer protocol server. The threat actor was able to access the server that contained RoSat flight mission files. Unbeknownst to the RoSat mission team, the threat actor changed values in the algorithms used by the system’s star tracker, thus making it point toward the sun and overheating. The team was able to

identify the issue and correct the satellite's positioning for what they thought was an accident, without knowing it was the result of cyber activities.

The foreign threat actor tried again months later—this time, changing the code for the altitude-control system. The satellite slewed out of control, pointing the X-ray imager toward the sun and irreparably damaging it completely.

Commercial Internet Service Provider Attack

On February 24, 2022, a multifaceted and deliberate cyberattack against a satellite network resulted in a partial interruption of the popular consumer-oriented satellite broadband server Viasat. The cyberattack impacted several thousand customers and tens of thousands of other fixed broadband customers across Europe. Ultimately, tens of thousands of modems that were previously online and active dropped off the network.

Subsequent investigation and forensic analysis identified a ground-based network intrusion by an attacker exploiting a misconfiguration in a virtual private network appliance to gain remote access to the trusted management segment of the satellite network. The attacker moved laterally through the trusted management network to a specific network segment used to manage and operate the satellites and then used the network's access to execute legitimate, targeted management commands on many residential modems simultaneously.

Juliana Suess, a research analyst and policy lead on space security at the defense think tank at the Royal United Services Institute, believes the cyberattack against the Viasat satellite system is a wake-up call to the space industry. The European Union, United Kingdom, and United States have linked the attack to Russia, prompting the U.S. National Security Agency to speak out about satellite security.

Conclusions

Cybersecurity in LEO satellite systems presents unique challenges and opportunities due to the specific characteristics of their orbits. The cyberspace landscape for LEO satellites is multifaceted, encompassing traditional cyberthreats such as unauthorized access, data interception, and denial-of-service attacks, together with satellite-specific vulnerabilities.

A fundamental challenge in LEO cybersecurity is managing its vast satellite constellations, where the sheer volume of satellites exponentially increases the attack surface and thereby amplifies the complexity of securing its network. Each satellite represents a potential vulnerability, underscoring the need for advanced authentication protocols and stringent data integrity safeguards across the entire constellation.

There is significant future potential in incorporating physical layer security methodologies and quantum cryptography for ultrasecure communication protocols and harnessing artificial intelligence for autonomous threat detection and mitigation. However, integrating such advanced technologies must be carefully calibrated to align with the inherent limitations of satellite systems, including power consumption, payload capacity, and computational constraints. Collaborative endeavors between satellite operators, cybersecurity professionals, and international regulatory organizations are critical in formulating comprehensive standards and frameworks that safeguard these assets while promoting continued innovation.

The security of LEO satellite systems will significantly influence their viability and reliability in delivering global connectivity, making it imperative to continually evolve cybersecurity strategies in this dynamic space environment. Lessons must be drawn from past cybersecurity incidents to shape a cyberspace future that is secure and resilient.

What's coming with 6G

6G networks will need to tackle numerous societal challenges, including demands for greater sustainability and economic resilience. They will also need to support opportunities for society and multiple industries to innovate on top of mobile networks. This includes granting access to a wider set of network services to enterprises and developers, while delivering new experiences to end consumers. New and evolved use cases will emerge that will demand more from networks. This will also lead to a substantial increase in traffic that networks will need to handle. Between 2024 and 2030 alone, total global mobile data traffic is forecast to grow by a factor of 2.5 – with a further acceleration expected in the era of 6G.

To make that possible, 6G will leverage new technological breakthroughs and apply new design concepts arising from today's rapid technology evolution including AI, compute hardware and cloud.

To this end, 6G will build on and expand further on the 5G capabilities and be better integrated into the rest of the ICT ecosystem of cloud platforms and applications.

Advanced use cases

6G will further advance the performance, service differentiation and guarantees of traditional use cases such as eMBB, FWA and IoT. An entirely new wave of use cases and devices will be made possible by 6G technologies, including synchronized distributed massive MIMO, extreme MIMO, non-terrestrial networks, integrated communication and sensing.. This is expected to include wide-area mixed reality, massive twinning, autonomous mobility, simultaneous location and mapping services, and more.

Evolution of the network platform

6G will take networks deeper into beyond-communication add-on services, making it possible for third-party developers to innovate with networks in entirely new ways. Information services such as network, positioning and sensing data, as well as new AI and compute services could all be exposed in the era of 6G. For CSPs, this will fuel new services and new relationships that can be monetized based on service differentiation.

AI-native networks

Through AI-native capabilities, 6G will take the industry significantly closer to fully autonomous network operations with zero human touch. Based on the principle of intelligence everywhere, AI workloads will be executed wherever it makes sense in the network based on a cost-benefit analysis. This will enhance resource allocation, minimize operational costs and create new monetization opportunities through AI-as-a-Service (AIaaS) models.

Sustainable by design

6G will be designed to make sustainable breakthroughs on all fronts: from improved network footprint to societal, economic and environmental benefits such as macro-economic gains, a narrower digital divide, and improvements to education, healthcare and other key sectors.

There are two sides to this journey: **sustainable 6G** (reducing 6G's impacts across the entire life cycle) and **6G for sustainability** (enabling use cases that create sustainable value for all).

Learn more about 6G and sustainability in our blog post: [Why 6G must put sustainability first](#)

5G vs. 6G: what to expect

Built on the foundations of 5G and 5G Advanced, 6G networks are expected to deliver much faster speeds than today's high-speed 5G networks. This includes the capability to provide several hundred gigabits per second (Gbps) and end-to-end sub-millisecond (ms) latency in specific scenarios and under certain conditions.

6G will not only be about speed. New functionalities will expand the possibilities of the network platform even further beyond communication services, such as information, AI, and compute services. This will grant developers deeper access to a much wider range of APIs. These APIs will build on capabilities from devices and servers, multiple network domains, and services beyond just connectivity. This will make it easier to meet the high demands of applications on communications and other capabilities from underlying infrastructure.

As a result, 6G will provide a much wider range of solutions to a larger customer base compared to today's 5G.

References

1. Rajesh Mehrotra (7 October 2011). "Regulation of Global Broadband Satellite Communications" (PDF). discussion paper. ITU. Retrieved 22 July 2012.
2. Patrick M. French (7 May 2009). "High Throughput Satellites (HTS) are pushing open the satellite market door" (PDF). guest column. Near Earth LLC. Archived from the original (PDF) on 3 December 2012. Retrieved 19 July 2012.
3. Jonathan Amos (22 October 2011). "Viasat broadband 'super-satellite' launches". news article. BBC. Retrieved 22 July 2012.
4. Giovanni Verlini (1 April 2011). "Next Generation of Satellite: High Capacity, High Potential". news article. Satellite Today. Archived from the original on 3 September 2020. Retrieved 19 July 2012.
5. Henry, Caleb (13 March 2018). "LEO and MEO broadband constellations mega source of

consternation". SpaceNews. Retrieved 25 March 2021.

6. Large LEO satellite constellations: Will it be different this time? McKinsey & Company, 4 May 2020, Accessed 25 march 2021
7. Low Earth Orbit satellites, improving latency OmniAccess, Accessed 29 October 2020
8. "The Bottom Line Archives Bottom Line". Northern Sky Research. Retrieved November 13, 2022.
9. "Satellite Mission Catalogue - WINDS". eoportal.org. Retrieved November 13, 2022.
10. "Kacific1 High Throughput Satellite". kacific.com. Retrieved November 13, 2022.
11. "Arianespace launches Eutelsat, ISRO satellites on first 2020 mission". SpaceNews. 2020-01-16. Retrieved 2020-06-06.
12. ESA. "Types of Orbits." https://www.esa.int/Enabling_Support/Space_Transportation/Types_of_orbits, accessed on 21 September 2024.
13. Space.com. "Starlink Satellites: Facts, Tracking and Impact on Astronomy." <https://www.space.com/spacex-starlink-satellites.html>, accessed on 13 September 2024.
14. Langbroek, M. "SpaceX Starlink Objects Train." SatTrackCam Leiden (b)log, <https://sattrackcam.blogspot.com/2019/05/wowowow-spectacular-view-of-spacex.html>, Screenshot, 29 May 2019.
15. National Aeronautics and Space Administration (NASA). "What Are SmallSats and CubeSats?" <https://www.nasa.gov/what-are-smallsats-and-cubesats>, accessed on 12 September 2024.
16. CubeSat. "CubeSat Design Specification." Rev. 14.1, CP-CDS-R14.1, CubeSat Program, Cal Poly, San Luis Obispo, CA, February 2022.
17. Johnson, N. L. "Medium Earth Orbits." IAC-10-A6.4.1, NASA, 2010.
18. NASA Science. "GOES Satellite Network." <https://science.nasa.gov/mission/goes/>, accessed on 22 September 2024.

19. Oakley, J. "Protecting the Final Frontier." Apress, 2020.
 20. NASA Earth Science Data Systems. "Remote Sensing." <https://www.earthdata.nasa.gov/learn/backgrounders/remote-sensing/>, accessed on 20 September 2024.
 21. Britannica. "Kármán Line." <https://www.britannica.com/science/Karman-line>, accessed on 18 September 2024.
 22. Wired. "Satellites Are Rife With Basic Security Flaws." <https://www.wired.com/story/satellites-basic-security-flaws/>, accessed on 28 September 2024.
 23. Straka, B. "Implementing a Microcontroller Watchdog With a Field-Programmable Gate Array (FPGA)." KSC-2013-091, Kennedy Space Center, April 2013.
 24. SPARTA. "Space Attack Research and Tactic Analysis (SPARTA)." Matrix, <https://sparta.aerospace.org>, accessed on 10 September 2024.
 25. U.S. Naval Institute. "ASAT Goes Cyber." <https://www.usni.org/magazines/proceedings/2021/february/asat-goes-cyber>, accessed on 15 September 2024.
 26. IMT Traffic Estimates for the Years 2020 to 2030. International Telecommunication Union (ITU) [(accessed on 25 October 2023)]. Available online: <https://www.itu.int/pub/r-rep-m.2370>.
-
27. Huseien G.F., Shah K.W. A Review on 5G Technology for Smart Energy Management and Smart Buildings in Singapore. *Energy AI*. 2022;7:100116. doi: 10.1016/j.egyai.2021.100116. [DOI] [Google Scholar]

28. 0.AI-Powered 6G Networks Will Reshape Digital Interactions. [(accessed on 29 October 2023)]. Available online: <https://www.technologyreview.com/2023/10/26/1082028/ai-powered-6g-networks-will-reshape-digital-interactions/>
29. 16.Techopedia. (n.d.-a). Interim Standard-95 (IS-95) [(accessed on 13 March 2023)]. Available online: <https://www.techopedia.com/definition/24248/interim-standard-95-is-95>.
30. 17.Moore K. From 1G to 5G: The Evolution of Mobile Communications. Mpirical, 11 August 2023. [(accessed on 30 October 2023)]. Available online: <https://www.mpirical.com/blog/the-evolution-of-mobile-communication>.
31. 19.Central Intelligence Agency. [(accessed on 30 October 2023)]; Available online: <https://www.cia.gov/the-world-factbook/field/telecommunication-systems>.
32. 20.Vergados D.D., Panoutsakopoulos A., Douligeris C. Group Registration with Distributed Databases for Location Tracking in 3G Wireless Networks. *Comput. Netw.* 2008;52:1521–1544. doi: 10.1016/j.comnet.2008.01.010. [DOI] [Google Scholar]
-
- 33.
34. 32.Kimachia K., Staff T., Clarke M., McQuarrie K., Millares L., Azhar A., Abbott B. 5G vs 6G: What's the Difference? TechRepublic, 23 February 2023. [(accessed on 1 November 2023)]. Available online: <https://www.techrepublic.com/article/5g-vs-6g/>
35. 35.Ruth C. How 6G Can Transform the World and Technology. IEEE Standards Association. 2023. [(accessed on 14 March 2024)]. Available online: <https://standards.ieee.org/beyond-standards/how-6g-can-transform-the-world-and-technology/>
36. 56.5G/6G Wireless Networks 5G/6G|Homeland Security. [(accessed on 2 November 2023)]; Available online: <https://www.dhs.gov/science-and-technology/5g6g>.

37. Morra J. Engineers Look to Adopt a More Sustainable Approach to Electronic Design. Electronic Design. 10 November 2023. [(accessed on 7 November 2023)]. Available online: <https://www.electronicdesign.com/resources/industry-insights/article/21275020/electronic-design-engineers-look-to-adopt-a-more-sustainable-approach-to-electronic-design>.
38. Edge Computing and Embedded Artificial Intelligence ECS SRIA. [(accessed on 23 February 2023)]. Available online: <https://ecssria.eu/2.1>.
39. Dong X.J. An Innovative Technology for 6G Communication Networks. Tech Xplore—Technology and Engineering News. 2022. [(accessed on 17 November 2023)]. Available online: <https://techxplore.com/news/2022-02-technology-6g-networks.html>.
40. Hall R. 5G and Gan: Understanding Sub-6ghz Massive Mimo Infrastructure. Embedded.com. 26 January 2021. [(accessed on 24 November 2023)]. Available online: <https://www.embedded.com/5g-and-gan-understanding-sub-6ghz-massive-mimo-infrastructure/>
41. US EPA Electric and Magnetic Fields from Power Lines. [(accessed on 25 November 2023)]; Available online: <https://www.epa.gov/radtown/electric-and-magnetic-fields-power-lines>.